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### Method for forming light pipe structure with high quantum efficiency

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#### Abstract

Various embodiments of the present disclosure are directed towards an integrated chip including an optical device within or on a semiconductor substrate. A light guide structure overlies the optical device. A first etch stop layer extends along first sidewalls and a lower surface of the light guide structure. A second etch stop layer overlies the first etch stop layer and extends along second sidewalls of the light guide structure.

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## Background/Summary

REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of U.S. application Ser. No. 17/406,297, filed on Aug. 19, 2021, which is a Divisional of U.S. application Ser. No. 16/405,027, filed on May 7, 2019 (now U.S. Pat. No. 11,121,162, issued on Sep. 14, 2021). The contents of the above-referenced patent applications are hereby incorporated by reference in their entirety.

## BACKGROUND

(1) Integrated circuits (IC) with image sensors are used in a wide range of modern day electronic devices, such as cameras and cell phones, for example. Complementary metal-oxide semiconductor (CMOS) devices have become popular IC image sensors. Compared to charge-coupled devices (CCD), CMOS image sensors are increasingly favored due to low power consumption, small size, fast data processing, a direct output of data, and low manufacturing cost. Some types of CMOS image sensors include front-side illuminated (FSI) image sensors and back-side illuminated (BSI) image sensors.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates a cross-sectional view of some embodiments of an image sensor comprising a light pipe structure overlying a photodetector, where a bottom surface of the light pipe structure extends below a bottommost conductive wire.

(3) FIGS. 2A and 2B illustrate cross-sectional views of various alternative embodiments of the image sensor of FIG. 1 in which the image sensor further comprises a bond pad overlying a stack of conductive layers.

(4) FIG. 3 illustrates a cross-sectional view of some embodiments of an image sensor comprising a light pipe structure over a photodetector and further comprising transistors adjacent to the photodetector, where the transistors are electrically coupled to overlying bond pads through an interconnect structure.

(5) FIGS. 4-15 illustrate a series of cross-sectional views of some embodiments of a method for forming an image sensor that comprises a light pipe structure over a photodetector and further comprises transistors adjacent to the photodetector.

(6) FIG. 16 illustrates a block diagram of some embodiments of the method of FIGS. 4-15.

### DETAILED DESCRIPTION

(7) The present disclosure provides many different embodiments, or examples, for implementing different features of this disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(8) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(9) A front-side illumination (FSI) image sensor may, for example, comprise light pipe structures

respectively overlying photodetectors. The light pipe structures are configured to guide incident radiation to underlying photodetector by total internal reflection (TIR), thereby increasing a quantum efficiency (QE) of the image sensor. The photodetectors are disposed within a semiconductor substrate. An interconnect structure overlies the semiconductor substrate. The interconnect structure comprises alternating stacks of wiring layers (e.g., horizontal routing) and via layers (e.g., vertical routing) disposed within an interconnect dielectric structure.

(10) The light pipe structures may be formed by performing an etch process (e.g., a dry etch process) through the interconnect dielectric structure after the interconnect structure is fully or substantially formed, thereby forming an opening above each photodetector. Subsequently, the light pipe structures may be formed in the openings. However, the etch may, for example, be performed by a plasma etch with a high power and hence a high electric field strength because the openings extend deep into the interconnect structure. This may cause electrons to accumulate on the semiconductor substrate, thereby increasing dark current and/or a number of white pixels present in the image sensor. Further, physical bombardment from ions of the plasma may damage the interconnect dielectric structure and/or damage a crystalline structure of the semiconductor substrate, thereby reducing a structural integrity of the interconnect dielectric structure and/or further increasing the dark current and/or the number of white pixels. Furthermore, the high power and a long duration of the etch process causes a high variation in heights of the light pipe structures. Therefore, the light pipe structures each extend from a top of the interconnect structure to different points above the semiconductor substrate, such that each light pipe structure may have a different height. It has been appreciated that the difference in height across the light pipe structures may cause non-uniformity across the photodetectors (e.g., a first photodetector may receive less incident radiation than an adjacent second photodetector) and may decrease an overall QE of the image sensor.

(11) Various embodiments of the present application are directed towards a method for forming a light pipe structure that increases the QE of the image sensor and decreases the dark current and/or the number of white pixels. In some embodiments, the method includes forming a lower interconnect portion of an interconnect structure over a photodetector and subsequently performing a first, low power etch into the lower interconnect portion to form an opening. A lower etch stop layer is formed lining the opening and has a U-shaped profile in the opening. An upper interconnect portion of the interconnect dielectric structure is formed over the lower etch stop layer and the lower interconnect portion. A second etch process is performed into the upper interconnect portion to form a light pipe opening overlying the photodetector. An upper etch stop layer is formed lining sidewalls of the light pipe opening. A wet etch process is performed to expand the light pipe opening and to remove excess material of the upper interconnect portion that overlies the lower etch stop layer. A light pipe structure is formed in the light pipe opening, such that a bottom surface of the light pipe structure is below a bottommost wiring layer of the interconnect structure. The use of the low power etch process to form the opening within which the lower etch stop layer is formed mitigates damage on a crystalline structure of the semiconductor substrate, thereby decreasing a dark current and/or a number of white pixels in the image sensor. Further, the wet etching process is able to form the light pipe opening while avoiding plasma damage that can occur during dry etching processes, thus mitigating the accumulation of electrons on the semiconductor substrate. Furthermore, the bottom surface of the light pipe structure extending below the bottommost wiring layer increases the QE and uniformity across the photodetectors in the image sensor.

(12) With reference to FIG. 1, a cross-sectional view of some embodiments of an image sensor **100** comprising a light pipe structure **115** overlying a photodetector **104** is provided in which a bottom surface **115b** of the light pipe structure **115** extends below a bottommost conductive wire **112**.

(13) The light pipe structure **115** overlies a photodetector **104** within a semiconductor substrate **102**. In some embodiments, the semiconductor substrate **102** may be, for example, a bulk substrate (e.g., a bulk silicon substrate), a silicon-on-insulator (SOI) substrate, or some other suitable

substrate having a first doping type (e.g., p-type). In further embodiments, the semiconductor substrate **102** may comprise three layers (e.g., silicon overlying an oxide and silicon underlying the oxide), such that a topmost layer in the three layers is epitaxial silicon with n-type doping. The photodetector **104** has a second doping type (e.g., n-type) opposite the first doping type. The photodetector **104** is configured to absorb incident radiation **120** (e.g., photons) and generate an electrical signal corresponding to the incident radiation **120**. In some embodiments, a depletion region forms (e.g., due to p-n junctions between the photodetector **104** and adjacent doped regions of the semiconductor substrate **102**) along a boundary of the photodetector **104**.

(14) A floating diffusion node **106** is disposed within the semiconductor substrate **102** laterally offset the photodetector **104**. The floating diffusion node **106** is within the semiconductor substrate **102** having the second doping type (e.g., n-type). A transfer transistor **110** is disposed over the semiconductor substrate **102** laterally between the photodetector **104** and the floating diffusion node **106**. The transfer transistor **110** may, for example, selectively form a conductive channel between the photodetector **104** and the floating diffusion node **106** to transfer accumulated charge in the photodetector **104** to the floating diffusion node **106**. The accumulated charge may, for example, arise from absorbing the incident radiation **120**.

(15) An interconnect structure **107** overlies the semiconductor substrate **102**. The interconnect structure **107** comprises conductive via(s) (e.g., a conductive contact **108**), conductive wire(s) (e.g., the bottommost conductive wire **112**), and dielectric layers and/or structures (e.g., a lower inter-level dielectric (ILD) structure **109**). The lower ILD structure **109** overlies the semiconductor substrate **102** and the transfer transistor **110**. The conductive contact **108** overlies the floating diffusion node **106** and extends through the lower ILD structure **109** to electrically couple the floating diffusion node **106** to the bottommost conductive wire **112**. In some embodiments, silicide may be disposed between the conductive contact **108** and the semiconductor substrate **102** (not shown). A lower etch stop structure **114** is disposed between the lower ILD structure **109** and an overlying upper ILD structure **116**. The lower etch stop structure **114** has a U-shape profile directly above the photodetector **104**. An upper etch stop structure **118** extends through the upper ILD structure **116** to the lower etch stop structure **114**.

(16) A light pipe structure **115** extends from a top of the interconnect structure **107** to below a bottom surface of the bottommost conductive wire **112**. In some embodiments, the light pipe structure **115** comprises a first dielectric material having a first refractive index (e.g., greater than 2.6), the lower etch stop structure **114** comprises a second dielectric material having a second refractive index (e.g., approximately 2.6), and the upper etch stop structure **118** comprises a third dielectric material having a third refractive index (e.g., approximately 2). In some embodiments, the first refractive index is greater than the second and third refractive indexes. By virtue of the first refractive index being greater than the second and third refractive indexes, the incident radiation **120** is confined to the light pipe structure **115** (e.g., due to total internal reflection) and is guided onto the photodetector **104**. In addition, because the second and third indexes are less than the first refractive index, a majority of the incident radiation **120** disposed on the interconnect structure **107** directly above the photodetector **104** is guided to the light pipe structure **115** and refracted toward the photodetector **104**.

(17) The bottom surface **115b** of the light pipe structure **115** extends below a top surface of the conductive contact **108** by a distance  $d_{sub.1}$ . In some embodiments, the distance  $d_{sub.1}$  is within a range of about 50 to 1500 Angstroms. In some embodiments, if the distance  $d_{sub.1}$  is small (e.g., less than about 50 Angstroms), then a height of the light pipe structure **115** is decreased. This, in part, may reduce incident radiation **120** disposed upon the photodetector **104**, thereby reducing a quantum efficiency (QE) of the image sensor **100**. In further embodiments, if the distance  $d_{sub.1}$  is large (e.g., greater than about 1500 Angstroms), then physical bombardment (e.g., from ions of a plasma used during a formation of the light pipe structure **115**) may damage the interconnect structure **107** and/or a crystalline structure of the semiconductor substrate **102**. This, in part, may

reduce a structural integrity of the interconnect structure **107**, increase a dark current in the photodetector **104**, and/or a number of white pixels in the image sensor **100**. In further embodiments, the lower ILD structure **109** comprises a fourth dielectric material (e.g., an oxide, such as silicon oxide) having a fourth refractive index (e.g., approximately 1.46) less than the first refractive index. In yet further embodiments, the upper ILD structure **116** comprises an oxide having a fifth dielectric material with a refractive index of approximately 1.3.

(18) In some embodiments, as seen in FIG. 1, the light pipe structure **115** extends continuously along substantially straight inner sidewalls of the upper etch stop structure **118**. In further embodiments, a top surface of the light pipe structure **115** is substantially aligned with a top surface of the upper etch stop structure **118**. An upper surface of the light pipe structure **115** is in direct contact with a bottom surface of the upper etch stop structure **118**. The lower etch stop structure **114** continuously extends along and cups outermost sidewalls and the bottom surface **115b** of the light pipe structure **115**. In some embodiments, the outermost sidewalls of the light pipe structure **115** are laterally between outer sidewalls of the photodetector **104**. The bottom surface **115b** of the light pipe structure **115** may, for example, be separated from a top surface of the photodetector **104** by a lower segment of the lower etch stop structure **114** and the lower ILD structure **109**. The light pipe structure **115** is recessed into the lower ILD structure **109**, such that the bottom surface **115b** of the light pipe structure **115** is below a top surface of the lower ILD structure **109**. The lower ILD structure **109** continuously extends along and cups outermost sidewalls and the bottom surface **115b** of the light pipe structure **115**. The transfer transistor **110** is disposed laterally between the light pipe structure **115** and the conductive contact **108**. The bottom surface **115b** of the light pipe structure **115** is disposed laterally above the top surface of the transfer transistor **110**. The light pipe structure **115** is laterally between an outer sidewall of the photodetector **104** and the transfer transistor **110**.

(19) With reference to FIG. 2A, a cross-sectional view of an image sensor **200a** according to some alternative embodiments of the image sensor **100** of FIG. 1 is provided, in which a bond pad **220** overlies the transfer transistor **110**.

(20) The image sensor **200a** includes the interconnect structure **107** overlying the semiconductor substrate **102**. A deep isolation structure **202** is disposed within the semiconductor substrate **102** adjacent to the photodetector **104**. In some embodiments, the deep isolation structure **202** may, for example, be a shallow trench isolation (STI) structure, a deep trench isolation (DTI) structure, or the like. In further embodiments, the deep isolation structure **202** comprises a dielectric material and/or extends from a front side surface **102f** of the semiconductor substrate **102** to a back side surface **102b** of the semiconductor substrate **102**. The deep isolation structure **202** is configured to electrically isolate the photodetector **104** from adjacent semiconductor devices (e.g., an adjacent photodetector) (not shown).

(21) The transfer transistor **110** is disposed between the floating diffusion node **106** and the photodetector **104**. In some embodiments, the transfer transistor **110** comprises a transfer gate electrode **234** overlying a transfer gate dielectric **232** and further comprises sidewall spacers **230** abutting the transfer gate electrode and dielectric **234**, **232**. In the aforementioned embodiment, a voltage may be applied to the transfer gate electrode **234** to control a transfer of accumulated charge (e.g., via absorbing incident radiation) in the photodetector **104** to the floating diffusion node **106**.

(22) The interconnect structure **107** overlies the front side surface **102f** of the semiconductor substrate **102**, such that the image sensor **200a** may, for example, be configured as a front-side illumination (FSI) image sensor. In some embodiments, the interconnect structure **107** comprises the conductive contact **108**, the bond pad **220**, an interconnect dielectric structure **201**, conductive wiring layers **210a-d**, and conductive vias **208** disposed within the interconnect dielectric structure **201**. The conductive contact **108** is disposed between a bottommost conductive wiring layer **210a** and the floating diffusion node **106**. The conductive vias **208** are disposed between the conductive

wiring layers **210a-d**. The interconnect dielectric structure **201** comprises a plurality of dielectric layers. The interconnect dielectric structure **201** includes the lower ILD structure **109**, the lower etch stop structure **114**, inter-wire buffer layers **204a-d**, the upper etch stop structure **118**, inter-metal dielectric (IMD) layers **206a-d**, and passivation layers **214a-b**. In some embodiments, a bottom surface of the upper etch stop structure **118** is disposed below an upper surface of a bottommost conductive wiring layer **210a**.

(23) In some embodiments, the lower ILD structure **109** may, for example, be or comprise one or more dielectric materials, such as an oxide, silicon oxide, a low-k dielectric, or the like, and/or may, for example, have a thickness within a range of about 2500 to 5000 Angstroms. As used herein, a low-k dielectric is a dielectric material that has a dielectric constant less than 3.9. In further embodiments, the lower etch stop structure **114** may, for example, be or comprise silicon carbide, or the like and/or may, for example, have a thickness within a range of about 200 to 500 Angstroms. In yet further embodiments, the inter-wire buffer layers **204a-c** may respectively, for example, be or comprise silicon carbide, or the like and/or may respectively, for example, have a thickness within a range of about 200 to 500 Angstroms. In some embodiments, the upper etch stop structure **118** and the inter-wire buffer layer **204d** may respectively, for example, be or comprise silicon nitride and/or may, for example, have a thickness within a range of about 250 to 750 Angstroms. In further embodiments, the IMD layers **206a-d** may respectively, for example, be or comprise an oxide, silicon oxide, a low-k dielectric, or the like and/or may respectively, for example, have a thickness within a range of about 1000 to 3000 Angstroms. In yet further embodiments, the passivation layers **214a-b** may respectively, for example, be or comprise an oxide, silicon oxide, a low-k dielectric, silicon nitride, or the like and/or may, for example, have a thickness within a range of about 500 to 2000 Angstroms. In some embodiments, the conductive contact **108**, the conductive vias **208**, and the conductive wiring layers **210a-d** may, for example, be or comprise a metal material, such as copper, tungsten, aluminum, or the like.

(24) The conductive vias **208** and the conductive wiring layers **210a-d** extend through the interconnect dielectric structure **201** and facilitate electrical coupling between underlying contact regions (e.g., the floating diffusion node **106**) and/or underlying semiconductor devices (e.g., the transfer transistor **110**) and overlying metal layers (e.g., the bond pad **220**). In some embodiments, the bond pad **220** may electrically couple an uppermost conductive wiring layer **210d** to semiconductor devices disposed on an external device (not shown). A solder bump **222** is disposed over the bond pad **220** to facilitate coupling between the bond pad **220** and an external I/O pin of an integrated chip package. A pad dielectric layer **218** is disposed between the bond pad **220** and the uppermost conductive wiring layer **210d**.

(25) The conductive contact **108** overlies the floating diffusion node **106** and may facilitate the transfer of the charge at the floating diffusion node **106** to overlying metal layers (e.g., the bond pad **220**). In some embodiments, a bottom surface of the conductive contact **108** is aligned with and directly contacts the front side surface **102f** of the semiconductor substrate **102**. In further embodiments, the bottom surface of the conductive contact **108** extends below the front side surface **102f** of the semiconductor substrate **102** (not shown). In yet further embodiments, the bottom surface of the conductive contact **108** is above the front side surface **102f** of the semiconductor substrate **102** and may be electrically coupled to the semiconductor substrate by way, for example, of a silicide, doped silicon, and/or polysilicon (not shown). The bottom surface **115b** of the light pipe structure **115** extends below a top surface of the conductive contact **108**. In some embodiments, a different conductive contact overlies the transfer gate electrode **234**, such that the bottom surface **115b** of the light pipe structure **115** extends below a top surface of the different conductive contact (not shown).

(26) The light pipe structure **115** extends through the interconnect dielectric structure **201** and terminates below a bottom surface of the bottommost conductive wiring layer **210a**. In some embodiments, the light pipe structure **115** has a first width  $W_{sub.1}$  greater than a second width

W.sub.2. Incident radiation disposed upon the interconnect structure **107** directly above the photodetector **104** is confined to the light pipe structure **115** (e.g., due to total internal reflection) and is guided to the photodetector **104**. The bottom surface **115b** of the light pipe structure **115** is disposed below the bottom surface of the bottommost conductive wiring layer **210a** by a distance d.sub.1. Thus, a thickness t.sub.1 of the lower ILD structure **109** between a bottom surface of the lower etch stop structure **114** and the front side surface **102f** of the semiconductor substrate **102** is reduced. This, in part, mitigates reflection and/or absorption of the incident radiation by the lower ILD structure **109**, thereby increasing a QE of the image sensor **200a**.

(27) The first width W.sub.1 is defined between opposing sidewalls **114s1**, **114s2** of the lower etch stop structure **114**. In some embodiments, the first width W.sub.1 is within a range of approximately 1 to 3.5 micrometers. The second width W.sub.2 is defined between opposing sidewalls **118s1**, **118s2** of the upper etch stop structure **118**. In some embodiments, the second width W.sub.2 is within a range of approximately 1 to 3 micrometers. In further embodiments, if the second width W.sub.2 is 1 micrometer or greater, then incident radiation disposed upon the photodetector **104** will be increased while reducing reflection of the incident radiation away from the photodetector **104** by the interconnect structure **107**. This, in part, will increase the overall QE of the image sensor **200a**. In yet further embodiments, if the second width W.sub.2 is 3 micrometers or less, then incident radiation disposed upon the photodetector **104** will be further increases while reducing a cost associated with forming the light pipe structure **115**. This, in part, will further increase the overall QE of the image sensor **200a**. In some embodiments, an absolute value of the difference between the first width W.sub.1 and the second width W.sub.2 (i.e., |W.sub.1-W.sub.2|) is less than 0.5 micrometers. In further embodiments, if the aforementioned difference is greater than, for example, 0.5 micrometers, then a structural integrity of the light pipe structure **115** may be reduced.

(28) Further, in some embodiments, the photodetector **104** may be in an array of photodetectors, such that the array comprises a plurality of photodetectors disposed in rows and columns. A light pipe structure directly overlies each photodetector in the array. The light pipe structures may each be configured as the light pipe structure **115**, such that a bottom surface of each light pipe structure extends through the interconnect dielectric structure **201** and terminates below a bottom surface of the bottommost conductive wiring layer **210a**. This, in part, may increase photodetector uniformity across the array (e.g., a first photodetector may receive about a same amount of incident radiation as an adjacent second photodetector), thereby increasing an overall QE of the image sensor **200a**.

(29) An anti-reflection layer **216** contacts the light pipe structure **115** and is configured to reduce the amount of incident radiation reflected by the interconnect structure **107**. In some embodiments, the anti-reflection layer **216** may, for example, be or comprise an oxide, a high-k dielectric, a nitride, or the like. A color filter **226** is disposed over the anti-reflection layer **216**. The color filter **226** is configured to transmit specific wavelengths of incident radiation while blocking other wavelengths of radiation. Further, a micro-lens **224** overlies the color filter **226** and is configured to focus the incident radiation towards the photodetector **104**.

(30) With reference to FIG. 2B, a cross-sectional view of an image sensor **200b** according to some alternative embodiments of the image sensor **200a** of FIG. 2A is provided, in which the lower etch stop structure **114** comprises a first lower etch stop layer **114a** underlying a second lower etch stop layer **114b**. The second lower etch stop layer **114b** has a U-shaped profile directly above the photodetector **104**. The lower etch stop structure **114** overlies a second bottommost conductive wiring layer **210b**. A bottom surface of the upper etch stop structure **118** is disposed above the top surface of the bottommost conductive wiring layer **210a**. The upper etch stop structure **118** has a first thickness t.sub.et1 between the top surface of the bottommost conductive wiring layer **210a** and the top surface of the lower etch stop structure **114**. In some embodiments, the first thickness t.sub.et1 is within a range of about 125 to 375 Angstroms. Further, the upper etch stop structure **118** has a second thickness t.sub.et2 above the bottom surface of the second bottommost conductive



wiring layer **210b**. In some embodiments, the second thickness  $t_{sub.et2}$  is within a range of about 250 to 750 Angstroms.

(31) With reference to FIG. 3, a cross-sectional view of an image sensor **300** according to some alternative embodiments of the image sensor **200a** of FIG. 2A is provided, in which a reset transistor **304** is disposed between the floating diffusion node **106** and a contact region **302**. In some embodiments, the reset transistor **304** comprise a reset gate electrode overlying a reset gate dielectric and further comprises sidewall spacers abutting the reset gate electrode and dielectric. In some embodiments, the contact region **302** is electrically coupled to a power supply (e.g., a DC power supply) supplying a reset voltage (e.g., 5 volts) by way of the interconnect structure **107**. In further embodiments, a reset gate voltage is applied to the reset gate electrode to apply the reset voltage to the floating diffusion node **106**. A conductive contact **108**, conductive wiring layers **210a-d**, conductive vias **208**, and a bond pad **220** overlie the contact region **302** and may, for example, be configured to electrically couple the contact region **302** to the power supply (not shown). A shallow isolation structure **306** is adjacent to the contact region **302** and is configured to electrically isolate the contact region **302** from adjacent semiconductor devices, adjacent contact regions, and/or adjacent source/drain regions (not shown). In some embodiments, the shallow isolation structure **306** may, for example, be a shallow trench isolation (STI) structure, a deep trench isolation (DTI) structure, or the like.

(32) With reference to FIG. 4, a cross-sectional view of an image sensor **400** according to some alternative embodiments of the image sensor **300** of FIG. 3 is provided, in which the lower etch stop structure **114** comprises a first lower etch stop layer **114a** underlying a second lower etch stop layer **114b**. The upper etch stop structure **118** comprises a first upper etch stop layer **118a** underlying a second upper etch stop layer **118b**. Further, an interconnect column **402** comprising conductive wiring layers **210a-d**, conductive vias **208**, a bond pad **220**, and a solder bump **222** is laterally offset from the transfer and reset transistors **110**, **304**. In some embodiments, the bond pad **220** is electrically coupled to the floating diffusion node **106** by way of the interconnect structure **107**. An inter-wire buffer layer **204d** is disposed between the bond pad **220** and the conductive vias **208**. In some embodiments, the inter-wire buffer layer **204d** comprises a same material as the upper etch stop structure **118**.

(33) FIGS. 5-15 illustrate cross-sectional views **500-1500** of some embodiments of a method of forming an image sensor device according to aspects of the present disclosure. Although the cross-sectional views **500-1500** shown in FIGS. 5-15 are described with reference to a method, it will be appreciated that the structures shown in FIGS. 5-15 are not limited to the method but rather may stand alone separate of the method. Although FIGS. 5-15 are described as a series of acts, it will be appreciated that these acts are not limiting in that the order of the acts can be altered in other embodiments, and the methods disclosed are also applicable to other structures. In other embodiments, some acts that are illustrated and/or described may be omitted in whole or in part. In some embodiments, FIGS. 5-15 may, for example, be employed to form the image sensor **400** of FIG. 4.

(34) As shown in cross-sectional view **500** of FIG. 5, a semiconductor substrate **102** is provided and a deep isolation structure **202** and a shallow isolation structure **306** are formed on a front side surface **102f** of the semiconductor substrate **102**. In some embodiments, the semiconductor substrate **102** may be, for example, a bulk substrate (e.g., a bulk silicon substrate), a silicon-on-insulator (SOI) substrate, or some other suitable substrate. In some embodiments, before forming the deep isolation structure **202** and the shallow isolation structure **306**, a first implant process is performed to dope the semiconductor substrate **102** with a first doping type (e.g., P-type). In some embodiments, a process for forming the deep isolation structure **202** and/or the shallow isolation structure **306** may comprise: 1) selectively etching the semiconductor substrate **102** to form a trench in the semiconductor substrate **102** that extends into the semiconductor substrate **102** from the front side surface **102f** of the semiconductor substrate **102**; and 2) filling (e.g., by chemical

vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), thermal oxidation, sputtering, etc.) the trench with a dielectric material. In further embodiments, the semiconductor substrate **102** is selectively etched by forming a masking layer (not shown) on the front side surface **102f** of the semiconductor substrate **102**, and subsequently exposing the semiconductor substrate **102** to an etchant configured to selectively remove unmasked portions of the semiconductor substrate **102**. In yet further embodiments, the dielectric material may comprise an oxide (e.g., silicon oxide), a nitride, or the like.

(35) Also shown in FIG. 5, a photodetector **104** is formed in the semiconductor substrate **102**. The photodetector **104** is a region of the semiconductor substrate **102** having a second doping type (e.g., N-type) opposite the first doping type. In some embodiments, the photodetector **104** may be formed by a selective ion implantation process that utilizes a masking layer (not shown) on the front side surface **102f** of the semiconductor substrate **102** to selectively implant ions into the semiconductor substrate **102**. Further, the transfer and reset transistors **110**, **304** are formed over the front side surface **102f** of the semiconductor substrate **102**. In some embodiments, a process for forming the transfer and/or reset transistors **110**, **304** comprises depositing and/or growing (e.g., by CVD, PVD, ALD, thermal oxidation, sputtering, etc.) a gate dielectric layer on the front side surface **102f** of the semiconductor substrate **102**. Next, a gate electrode layer may be deposited on the gate dielectric layer. Subsequently, the gate dielectric and electrode layers are patterned (e.g., by a photolithography/etching process) to form gate dielectrics and gate electrodes, respectively. Further, spacers may, for example, be formed by: 1) depositing (e.g., by CVD, PVD, ALD, sputtering, etc.) a spacer layer over the front side surface **102f** of the semiconductor substrate **102**, the gate dielectrics, and the gate electrodes; and 2) etching back the spacer layer to remove the spacer layer from horizontal surfaces. In further embodiments, the spacer layer may comprise a nitride, an oxide, or some other dielectric. In further embodiments, the gate electrode layer may comprise, for example, polysilicon, aluminum or the like. In yet further embodiments, the gate dielectric layer may comprise, for example, an oxide, a high-k dielectric, or the like.

(36) Also shown in FIG. 5, the contact region **302** and the floating diffusion node **106** are formed in the semiconductor substrate **102** on opposing sides of the reset transistor **304**.

(37) In some embodiments, the contact region **302** and the floating diffusion node **106** are regions of the semiconductor substrate **102** having the second doping type (e.g., N-type). In some embodiments, the contact region **302** and the floating diffusion node **106** may be formed by a selective ion implantation process that utilizes a masking layer (not shown) disposed on the front side surface **102f** of the semiconductor substrate **102** to selectively implant n-type dopants (e.g., phosphorus) into the semiconductor substrate **102**. Further, a lower inter-level dielectric (ILD) structure **109** is formed over the front side surface **102f** of the semiconductor substrate **102**. In some embodiments, the lower ILD structure **109** may, for example, be or comprise one or more dielectric materials, such as an oxide, silicon oxide, a low-k dielectric, or the like, and/or may, for example, have a thickness within a range of about 2500 to 5000 Angstroms. Formation of the lower ILD structure **109** may, for example, comprise deposition followed by a planarization to flatten a top surface of the lower ILD structure **109**.

(38) Furthermore, conductive contacts **108** are formed within the lower ILD structure **109** over the contact region **302** and the floating diffusion node **106**. In some embodiments, the conductive contacts **108** are formed by a single damascene process. In some embodiments, conductive contacts **108** are formed over the transfer and reset transistors **110**, **304** (not shown). In some embodiments, the single damascene process comprises patterning a dielectric layer and/or structure (e.g., the lower ILD structure **109**) with openings for a single layer of conductive features (e.g., layer of contacts, vias, or wires), and filling the openings with conductive materials to form the single layer of conductive features. In further embodiments, a planarization process (e.g., a chemical mechanical planarization (CMP) process) is performed such that a top surface of the ILD structure **109** is aligned with top surfaces of the conductive contacts **108**. Additionally, an inter-wire buffer

layer **204a** is formed over the lower ILD structure **109** and an inter-metal dielectric (IMD) layer **206a** is formed over the inter-wire buffer layer **204a**. In some embodiments, the formation of the ILD structure **109**, the inter-wire buffer layer **204a**, and/or the IMD layer **206a** may, for example, utilize CVD, PVD, ALD, and/or sputtering. In yet further embodiments, the inter-wire buffer layer **204a** may, for example, be or comprise silicon carbide, or the like and/or may, for example, have a thickness within a range of about 200 to 500 Angstroms. In further embodiments, the IMD layer **206a** may, for example, be or comprise an oxide, silicon oxide, a low-k dielectric, or the like and/or may, for example, have a thickness within a range of about 1000 to 3000 Angstroms.

(39) As shown in cross-sectional view **600** of FIG. **6**, a bottommost conductive wiring layer **210a** is formed over the lower ILD structure **109**. In some embodiments, the bottommost conductive wiring layer **210a** is formed, for example, by a single damascene process. In some embodiments, the bottommost conductive wiring layer **210a** may, for example, be or comprise a metal material, such as copper, tungsten, aluminum, or the like.

(40) As shown in cross-sectional view **700** of FIG. **7**, a first lower etch stop layer **114a** is formed over the IMD layer **206a**. Subsequently, an etching process is performed to define a first opening **702** above the photodetector **104**. In some embodiments, the etching process includes forming a masking layer over the first lower etch stop layer **114a**, selectively exposing the layers below the masking layer to one or more etchants, and removing the masking layer (not shown). In some embodiments, the etching process removes a portion of the ILD structure **109**, such that an upper surface of the ILD structure **109** is recessed below a bottom surface of the bottommost conductive wiring layer **210a** by a recess distance  $r_{\text{sub.1}}$ . In some embodiments, the recess distance  $r_{\text{sub.1}}$  is, for example, within a range of approximately 500 to 2000 Angstroms. A thickness  $t_{\text{sub.1}}$  of the ILD structure **109** is defined between the upper surface of the ILD structure **109** and a top surface of the semiconductor substrate **102**. In some embodiments, the thickness  $t_{\text{sub.1}}$  is, for example, within a range of approximately 500 to 2000 Angstroms. A reduction of the thickness  $t_{\text{sub.1}}$  of the ILD structure **109** over the photodetector **104** may, for example, increase an amount of incident radiation the photodetector **104** may receive, thereby increasing a QE of the photodetector **104**.

(41) In some embodiments, the etching process utilized to form the first opening **702** may, for example, be a low power dry etch process (e.g., the low power etch process may have a low power within a range of about 200 to 400 Watts (W)). Further, the one or more etchants utilized in the low power etch process may, for example, be or comprise a fluorine base chemical (e.g., perfluorocyclobutane ( $\text{C}_{\text{sub.4}}\text{F}_{\text{sub.8}}$ ), hexafluorocyclobutene ( $\text{C}_{\text{sub.4}}\text{F}_{\text{sub.6}}$ )), argon, helium, and/or the like. The use of the low power dry etch process and the one or more etchants mitigates damage to the semiconductor substrate **102** and/or mitigates an accumulation of electrons on the semiconductor substrate **102**. This, in part, decreases the presence of a dark current and/or a white pixel in the photodetector **104**.

(42) As shown in cross-sectional view **800** of FIG. **8**, a second lower etch stop layer **114b** is formed over the first lower etch stop layer **114a** and within the first opening (**702** of FIG. **7**). In some embodiments, the first and second lower etch stop layers **114a**, **114b** define a lower etch stop structure **114**. In further embodiments, the first and second lower etch stop layers **114a**, **114b** respectively comprise silicon carbide and are respectively formed to a thickness within a range of about 200 to 500 Angstroms. The second lower etch stop layer **114b** has a U-shaped segment **114bu** directly overlying the photodetector **104**.

(43) As shown in cross-sectional view **900** of FIG. **9**, a second IMD layer **206b** is formed over the lower etch stop structure **114**. In some embodiments, the second IMD layer **206b** comprises a same material as the IMD layer **206a**. The second IMD layer **206b** fills the U-shaped segment (**114bu** of FIG. **8**) of the second lower etch stop layer **114b**. Formation of the second IMD layer **206b** may, for example, comprise deposition followed by a planarization to flatten a top surface of the second IMD layer **206b**.

(44) As shown in cross-sectional view **1000** of FIG. **10**, conductive layers and dielectric layers of

an interconnect structure **107** are formed over the semiconductor substrate **102**. An interconnect dielectric structure **201** comprises inter-wire buffer layers **204a-d**, IMD layers **206a-d**, passivation layers **214a-b**, and a first upper etch stop layer **118a**. In some embodiments, the dielectric layers within the interconnect dielectric structure **201** respectively may, for example, be formed by CVD, PVD, ALD, thermal oxidation, sputtering, etc. In some embodiments, the first upper etch stop layer **118a** may, for example, be or comprise silicon nitride and/or may, for example, be formed to a thickness within a range of about 250 to 750 Angstroms. The conductive layers of the interconnect structure **107** comprise conductive wiring layers **210a-d**, and conductive vias **208** formed within the interconnect dielectric structure **201**. In some embodiments, the conductive wiring layers **210a-d** and/or the conductive vias **208** may, for example, be formed by a single or a dual damascene process. In some embodiments, the dual damascene process comprises depositing a dielectric layer, patterning the dielectric layer with openings for two layers of conductive features (e.g., a layer of vias and a layer of wires), and filling the openings with conductive material to form the two layers of conductive features. The dielectric layer may, for example, correspond to one of the layers in the interconnect dielectric structure **201**. In further embodiments, a bond pad **220** is formed over an interconnect column **402**. A pad dielectric layer **218** is formed between the bond pad **220** and an uppermost conductive wiring layer **210d**. A solder bump **222** is formed over the bond pad **220**. The interconnect column **402** is laterally offset from the transfer and reset transistors **110**, **304**. In some embodiments, the bond pad **220**, the pad dielectric layer **218**, and solder bump **222** may be formed above the transfer and/or reset transistors **110**, **304** (not shown).

(45) In some embodiments, a process for forming the bond pad **220** and the pad dielectric layer **218** may include: 1) patterning the passivation layers **214a-b** and the first upper etch stop layer **118a** to define a bond pad opening (not shown); 2) lining the bond pad opening with the pad dielectric layer **218**; 3) selectively etching (e.g., by a masking layer (not shown)) the pad dielectric layer **218** to form openings that expose an upper surface of the uppermost conductive wiring layer **210d** and remove the pad dielectric layer **218** from sidewalls of the bond pad opening; 4) selectively forming the bond pad **220** over the pad dielectric layer **218**, such that sidewalls of the bond pad **220** are laterally offset sidewalls of the bond pad opening by a non-zero distance. In some embodiments, step 4) in the aforementioned process may be achieved by forming the bond pad **220** over the pad dielectric layer **218** and subsequently selectively etching the bond pad **220** by a masking layer (not shown).

(46) As shown in cross-sectional view **1100** of FIG. **11**, an etching process is performed to define a second opening **1102** in the interconnect dielectric structure **201** above the photodetector **104**. In some embodiments, the etching process includes forming a masking layer over the first upper etch stop layer **118a**, selectively exposing layers below the masking layer to one or more etchants, and subsequently removing the masking layer (not shown). In some embodiments, the etching process removes a portion of the second IMD layer **206b** and a portion of the second lower etch stop layer **114b**, such that an upper surface of the second IMD layer **206b** is recessed below a top surface of the bottommost conductive wiring layer **210a** by a recess distance  $r_2$ . In some embodiments, the recess distance  $r_2$  is within a range of approximately 100 to 400 Angstroms.

(47) In some embodiments, the etching process utilized to form the second opening **1102** may, for example, be a high power dry etch process (e.g., the high power dry etch process may have a high power within a range of about 1000 to 2500 W). Further, the one or more etchants utilized in the high power dry etch process may, for example, be or comprise a fluorine base chemical (e.g., perfluorocyclobutane (C<sub>sub</sub>4F<sub>sub</sub>8), hexafluorocyclobutene (C<sub>sub</sub>4F<sub>sub</sub>6)), argon, helium, and/or the like. A thickness and configuration of the lower etch stop structure **114** and/or the second IMD layer **206b** mitigates an accumulation of electrons on the semiconductor substrate **102**, and/or a damage on a crystalline structure of the semiconductor substrate **102** during the high power etch process. This, in part, decreases the presence of a dark current and/or a white pixel in the photodetector **104**.

(48) As shown in cross-sectional view **1200** of FIG. **12**, a second upper etch stop layer **118b** is formed over the first upper etch stop layer **118a** and within the second opening (**1102** of FIG. **11**). In some embodiments, the first and second upper etch stop layers **118a**, **118b** define an upper etch stop structure **118**. In further embodiments, the first and second upper etch stop layers **118a**, **118b** respectively comprise silicon nitride and/or are respectively formed to a thickness within a range of about 250 to 750 Angstroms. The second upper etch stop layer **118b** has a U-shaped segment **118bu** directly overlying the photodetector **104**.

(49) As shown in cross-sectional view **1300** of FIG. **13**, an etching process is performed to remove a bottom segment of the upper etch stop structure **118** and to expose the second IMD layer **206b**. In some embodiments, the etching process includes forming a masking layer over the second upper etch stop layer **118b**, selectively exposing the second upper etch stop layer **118b** to one or more etchants according to the masking layer, and subsequently removing the masking layer (not shown). In alternative embodiments, the etching process comprises an etch back and hence does not rely upon a masking layer. The etching process exposes an upper surface of a lower segment **206b1** of the second IMD layer **206b**.

(50) In some embodiments, the etching process utilized to remove the lower segment of the U-shaped segment (**118bu** of FIG. **12**) may, for example, be a low power dry etch process (e.g., the low power dry etch process may have a low power within a range of about 200 to 400 W). Further the one or more etchants utilized in the lower power etch process may, for example, be or comprise fluoromethane (e.g., CHF, CHF.sub.3, etc.), or the like. The use of the low power dry etch process and the one or more etchants mitigates an accumulation of electrons on the semiconductor substrate **102**. This, in part, decreases the presence of a dark current and/or a white pixel in the photodetector **104**. Further, the second upper etch stop layer **118b** may be configured to protect the interconnect dielectric structure **201** from the low power dry etch process, thereby mitigating physical bombardment from ions of the one or more etchants on the dielectric layers in the interconnect dielectric structure **201**.

(51) As shown in cross-sectional view **1400** of FIG. **14**, an etching process is performed to remove the lower segment (**206b1** of FIG. **13**) of the second IMD layer **206b**, thereby defining a light pipe opening **1402**. In some embodiments, the etching process is a wet etch process. Further, the wet etch process may, for example, utilize one or more etchants (e.g., hydrogen fluoride). In some embodiments, the second upper etch stop layer **118b** covers the solder bump **222**, such that the second upper etch stop layer **118b** covers and protects the solder bump **222** and the bond pad **220** from the one or more etchants during the wet etch process. Further, the wet etch process is able to form the light pipe opening **1402** while avoiding plasma damage that may occur during a dry etching process, thereby further mitigating the accumulation of electrons on the semiconductor substrate **102**.

(52) As shown in cross-sectional view **1500** of FIG. **15**, a light pipe structure **115** is formed within the light pipe opening (**1402** of FIG. **14**). A bottom surface **115b** of the light pipe structure **115** extends below a bottom surface of the bottommost conductive wiring layer **210a** by a distance d.sub.1. In some embodiments, the distance d.sub.1 is within a range of about 50 to 1500 Angstroms. In some embodiments, if the distance d.sub.1 is small (e.g., less than about 50 Angstroms), then a height of the light pipe structure **115** is decreased. This, in part, may reduce incident radiation **120** disposed upon the photodetector **104**, thereby reducing a quantum efficiency (QE) of the image sensor **100**. In further embodiments, if the distance d.sub.1 is large (e.g., greater than about 1500 Angstroms), then physical bombardment (e.g., from ions of a plasma used during a formation of the light pipe structure **115**) may damage the interconnect structure **107** and/or a crystalline structure of the semiconductor substrate **102**. This, in part, may reduce a structural integrity of the interconnect structure **107**, increase a dark current in the photodetector **104**, and/or a number of white pixels in the image sensor **100**. In some embodiments, the low power dry etch process and the lower etch stop structure **114** facilitates a high degree of control over a height of the

light pipe structure **115**. Further, this process may be completed for a plurality of light pipe structures **115**, such that each light pipe structure **115** may, for example, have approximately a same height.

(53) Also shown in FIG. **15**, after forming the light pipe structure **115**, a removal process is performed to remove the second upper etch stop layer **118b** over and/or around the bond pad **220**. In some embodiments, the removal process includes forming a masking layer over the second upper etch stop layer **118b**, performing an etching process according to the masking layer, and subsequently removing the masking layer (not shown). In further embodiments, the removal process includes reducing a thickness of the second upper etch stop layer **118b** overlying the passivation layers **214a-b**. Further, an anti-reflection layer **216** is formed over the light pipe structure **115**. A color filter **226** (e.g., a red color filter, a blue color filter, a green color filter, etc.) is formed over the anti-reflection layer **216**. Further, a micro-lens **224** is formed over the color filter **226**.

(54) FIG. **16** illustrates a method **1600** of forming an image sensor device that includes a light pipe structure over a photodetector and transistors adjacent to the photodetector according to the present disclosure. Although the method **1600** is illustrated and/or described as a series of acts or events, it will be appreciated that the method is not limited to the illustrated ordering or acts. Thus, in some embodiments, the acts may be carried out in different orders than illustrated, and/or may be carried out concurrently. Further, in some embodiments, the illustrated acts or events may be subdivided into multiple acts or events, which may be carried out at separate times or concurrently with other acts or sub-acts. In some embodiments, some illustrated acts or events may be omitted, and other un-illustrated acts or events may be included.

(55) At act **1602**, a photodetector is formed in a semiconductor substrate and a plurality of pixel devices are formed over the semiconductor substrate. FIG. **5** illustrates a cross-sectional view **500** corresponding to some embodiments of act **1602**.

(56) At act **1604**, an inter-layer dielectric (ILD) structure is formed over the semiconductor substrate. FIG. **5** illustrates a cross-sectional view **500** corresponding to some embodiments of act **1604**.

(57) At act **1606**, conductive contacts are formed over the semiconductor substrate, such that the conductive contacts extend through the ILD structure to the pixel devices. FIG. **5** illustrates a cross-sectional view **500** corresponding to some embodiments of act **1606**.

(58) At act **1608**, a first etch stop layer and a first inter-metal dielectric (IMD) layer are formed over the ILD structure. FIG. **5** illustrates a cross-sectional view **500** corresponding to some embodiments of act **1608**.

(59) At act **1610**, a bottommost conductive wiring layer is formed over the conductive contacts, recessed into the first IMD layer. FIG. **6** illustrates a cross-sectional view **600** corresponding to some embodiments of act **1610**.

(60) At act **1612**, a second etch stop layer is formed over the bottommost conductive wiring layer and first IMD layer. FIG. **7** illustrates a cross-sectional view **700** corresponding to some embodiments of act **1612**.

(61) At act **1614**, a first etch process is performed to define a first opening above the photodetector, such that the first etch process removes a portion of the ILD structure. FIG. **7** illustrates a cross-sectional view **700** corresponding to some embodiments of act **1614**.

(62) At act **1616**, another etch stop layer having the same material as the second etch stop layer is formed in the first opening, such that the second etch stop layer comprises the another etch stop layer and has a U-shape above the photodetector. FIG. **8** illustrates a cross-sectional view **800** corresponding to some embodiments of act **1616**.

(63) At act **1618**, dielectric layers, etch stop layers, conductive vias, and conductive wiring layers are formed over the semiconductor substrate. FIGS. **9** and **10** illustrate cross-sectional views **900** and **1000** corresponding to some embodiments of act **1618**.

(64) At act **1620**, a second etch process is performed to form a second opening directly above the U-shape of the second etch stop layer. FIG. **11** illustrates a cross-sectional view **1100** corresponding to some embodiments of act **1620**.

(65) At act **1622**, a topmost etch stop layer is formed in the second opening. FIG. **12** illustrates a cross-sectional view **1200** corresponding to some embodiments of act **1622**.

(66) At act **1624**, a removal process is performed to expose an upper surface of the second etch stop layer and define a light pipe opening. FIGS. **13** and **14** illustrate cross-sectional views **1300** and **1400** corresponding to some embodiments of act **1624**.

(67) At act **1626**, a light pipe structure is formed in the light pipe opening. FIG. **15** illustrates a cross-sectional view **1500** corresponding to some embodiments of act **1626**.

(68) At act **1628**, an anti-reflection layer, a color filter, and a micro-lens is formed over the light pipe structure. FIG. **15** illustrates a cross-sectional view **1500** corresponding to some embodiments of act **1628**.

(69) Accordingly, in some embodiments, the present application provides an image sensor device that includes a light pipe structure over a photodetector and transistors adjacent to the photodetector. A bottom surface of the light pipe structure is below a bottom surface of a bottommost conductive wire in an interconnect structure.

(70) In some embodiments, the present application provides an image sensor, including a semiconductor substrate; a photodetector disposed within the semiconductor substrate; a gate electrode overlying the semiconductor substrate and bordering the photodetector; an inter-level dielectric (ILD) layer overlying the semiconductor substrate; a conductive contact within the ILD layer, wherein a bottom surface of the conductive contact is below a top surface of the gate electrode; and a light pipe structure overlying the photodetector, wherein a bottom surface of the light pipe structure is recessed below a top surface of the conductive contact.

(71) In some embodiments, the present application provides a complementary metal-oxide-semiconductor (CMOS) image sensor, including a substrate; a photodetector disposed within the substrate; an interconnect structure comprising an inter-level dielectric (ILD) layer, an inter-metal dielectric (IMD) structure overlying the inter-level dielectric (ILD) layer, and conductive wires within the IMD structure, wherein the interconnect structure overlies the substrate and is selectively electrically coupled to the photodetector; and a light pipe structure overlying the photodetector and continuously extending from a top of the IMD structure to a point below a bottom surface of a bottommost conductive wire of the interconnect structure.

(72) In some embodiments, the present application provides a method for forming an image sensor, the method including forming a photodetector in a semiconductor substrate; forming a lower interconnect portion of an interconnect structure over the photodetector; performing a first removal process to define a first opening overlying the photodetector in the lower interconnect portion; forming a lower etch stop layer lining the first opening, wherein the lower etch stop layer has a U-shape in the first opening; forming an upper interconnect portion of the interconnect structure over the lower etch stop layer; and forming a light pipe structure overlying the photodetector, wherein the U-shape of the lower etch stop layer extends continuously along sidewalls and a bottom surface of the light pipe structure.

(73) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

## Claims

1. An integrated chip, comprising: an optical device disposed within or on a semiconductor substrate; a light guide structure overlying the optical device and comprising first outer sidewalls and a bottom surface continuously extending between the first outer sidewalls; a first etch stop layer continuously extending along the first outer sidewalls and the bottom surface of the light guide structure; and a second etch stop layer over the first etch stop layer and extending along second outer sidewalls of the light guide structure.
2. The integrated chip of claim 1, wherein a height of the second etch stop layer is greater than a height of the first etch stop layer.
3. The integrated chip of claim 1, further comprising: a conductive wire over the semiconductor substrate and laterally offset from the light guide structure, wherein the first etch stop layer continuously laterally extends from a top surface of the conductive wire to the first outer sidewalls of the light guide structure.
4. The integrated chip of claim 1, wherein the second etch stop layer contacts opposing inner sidewalls of the first etch stop layer.
5. The integrated chip of claim 1, wherein the second etch stop layer continuously vertically extends from a top surface of the first etch stop layer to a top surface of the light guide structure.
6. The integrated chip of claim 1, further comprising: an interconnect dielectric structure over the optical device, wherein the light guide structure is disposed within the interconnect dielectric structure; and a bond pad disposed within the interconnect dielectric structure and laterally offset from the light guide structure, wherein a top surface of the light guide structure is aligned with or vertically below a top surface of the bond pad.
7. The integrated chip of claim 6, further comprising: a bottommost conductive wire disposed within the interconnect dielectric structure and directly underlying the bond pad, wherein the bottom surface of the light guide structure is disposed below a bottom surface of the bottommost conductive wire by a first distance, wherein the bottom surface of the light guide structure is vertically offset from a top surface of the semiconductor substrate by a second distance greater than the first distance.
8. The integrated chip of claim 6, wherein the second etch stop layer is disposed on opposing sides of the bond pad.
9. The integrated chip of claim 1, further comprising: a color filter overlying the light guide structure; and a micro-lens disposed on the color filter.
10. An integrated chip, comprising: an optical device disposed within or on a semiconductor substrate; an interconnect structure disposed over the semiconductor substrate; and a light guide structure disposed in the interconnect structure and directly overlying the optical device, wherein the light guide structure comprises a bottom light guide segment and a top light guide segment over a top surface of the bottom light guide segment, wherein opposing sidewalls of the top light guide segment are spaced between opposing sidewalls of the bottom light guide segment.
11. The integrated chip of claim 10, wherein the interconnect structure comprises a conductive contact over the semiconductor substrate and a conductive wire disposed on the conductive contact, wherein the top surface of the bottom light guide segment is vertically below a top surface of the conductive wire.
12. The integrated chip of claim 11, further comprising: a first etch stop layer disposed along the opposing sidewalls of the bottom light guide segment, wherein the first etch stop layer continuously laterally extends from the top surface of the conductive wire to the bottom light guide segment.
13. The integrated chip of claim 12, further comprising: a lower etch stop layer over the conductive contact and below a top surface of the first etch stop layer, wherein the lower etch stop layer



contacts outer sidewalls of the first etch stop layer.

14. The integrated chip of claim 10, further comprising: a first etch stop layer disposed along the opposing sidewalls of the bottom light guide segment, wherein a top surface of the first etch stop layer is below a horizontal line extending through a middle of a height of the top light guide segment.

15. The integrated chip of claim 14, further comprising: a second etch stop layer extending along the opposing sidewalls of the top light guide segment, wherein the second etch stop layer contacts the top surface of the bottom light guide segment.

16. The integrated chip of claim 10, wherein a height of the top light guide segment is greater than a height of the bottom light guide segment.

17. An integrated chip, comprising: an optical device disposed within or on a semiconductor substrate; a dielectric structure over the semiconductor substrate, wherein the dielectric structure comprises first opposing sidewalls and a lower surface over the optical device, wherein the dielectric structure comprises second opposing sidewalls between the first opposing sidewalls and the lower surface of the dielectric structure; a first dielectric protection layer disposed along the lower surface of the dielectric structure; and a second dielectric protection layer disposed along the first opposing sidewalls of the dielectric structure and over the first dielectric protection layer, wherein a height of the second dielectric protection layer is greater than a height of the first dielectric protection layer, wherein the first dielectric protection layer extends along the second opposing sidewalls of the dielectric structure to a lower surface of the second dielectric protection layer.

18. The integrated chip of claim 17, wherein a first distance between inner opposing sidewalls of the first dielectric protection layer is greater than a second distance between inner opposing sidewalls of the second dielectric protection layer.

19. The integrated chip of claim 17, wherein the second dielectric protection layer continuously vertically extends from an upper surface of the first dielectric protection layer to a point aligned with a top surface of the dielectric structure.

20. The integrated chip of claim 17, wherein in a cross-sectional view a bottom surface of the second dielectric protection layer is discontinuous in a region between inner opposing sidewalls of the second dielectric protection layer, wherein the first dielectric protection layer has a planar surface under the region.

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